

Subject: Cryptography and network security
Name: Atchaya Chandran R
Reg no: 192521394

ASSIGNMENT: CRYPTOGRAPHY AND NETWORK SECURITY

PART I: EVENT COMMUNICATION SECURITY (QUESTION 1)

1. Introduction to Classical Cryptography and Shift Ciphers

The foundation of modern cryptography lies deep within classical systems. Among the earliest known geometric and algebraic methods of masking plaintext is the Shift Cipher, colloquially recognized as the Caesar Cipher. Named after Julius Caesar, who historically utilized it to protect critical military correspondence, this technique operates on a monoalphabetic substitution principle. In a monoalphabetic system, a single, fixed cryptographic key governs the transformation of each character throughout the entire duration of the message. This ensures that every individual letter maps uniquely to exactly one corresponding ciphertext character based on a fixed spatial offset.

In the modern context of Network Security, classical ciphers are considered computationally insecure and structurally obsolete. However, they serve as crucial pedagogical primitives for understanding symmetric key distribution, mathematical modular arithmetic, and foundational cryptanalysis. This section evaluates a scenario where a student organization attempts to secure its internal logistics by encrypting the plaintext phrase: "MEET AT NOON" using a fixed shift pattern configuration, exploring its mechanics, structural behavior, and ultimate architectural vulnerabilities.

2. Mathematical Formulation of the Shift Cipher

To analyze the transformation objectively, we map the standard English alphabet (A through Z) to a finite set of sequential integers within the ring of integers modulo 26, denoted mathematically as $\mathbb{Z}_{26}$. The mapping configuration is explicitly structured as follows:

A=0, B=1, C=2, D=3, E=4, F=5, G=6, H=7, I=8, J=9, K=10, L=11, M=12,
N=13, O=14, P=15, Q=16, R=17, S=18, T=19, U=20, V=21, W=22, X=23, Y=24, Z=25

Let P represent the numerical value of a plaintext letter, C represent the numerical value of the corresponding ciphertext letter, and K represent the secret key parameter (shift value). The encryption function E and decryption function D are defined algebraically using modular congruence:

$$E(P) = (P + K) \bmod 26$$

$$D(C) = (C - K) \bmod 26$$

In this specific scenario, the student organization has selected a fixed shift pattern where $K = 4$. This implies that every letter in the alphabet will be systematically advanced by four positions forward along the modular ring.

3. Step-by-Step Encryption Process ($K = 4$)

Below is the granular, character-by-character transformation matrix utilized to reconstruct how the encrypted message appears to an unauthorized eavesdropper monitoring the network. Non-alphabetical spaces between words are structurally preserved to maintain message parsing context, though in highly secure deployments, spacing is typically omitted or filled with null values to prevent structural word-length analysis.

Character 1: 'M'

Plaintext Value (P): 12

Calculation: $E(12) = (12 + 4) \bmod 26 = 16 \bmod 26$

Ciphertext Letter: **Q**

Character 2: 'E'

Plaintext Value (P): 4

Calculation: $E(4) = (4 + 4) \bmod 26 = 8 \bmod 26$

Ciphertext Letter: **I**

Character 3: 'E'

Plaintext Value (P): 4

Calculation: $E(4) = (4 + 4) \bmod 26 = 8 \bmod 26$

Ciphertext Letter: **I**

Character 4: 'T'

Plaintext Value (P): 19

Calculation: $E(19) = (19 + 4) \bmod 26 = 23 \bmod 26$

Ciphertext Letter: **X**

Character 5: 'A'

Plaintext Value ()P0

Calculation: $E(0)=(0+4) \bmod 26 = 4 \bmod 26$

Ciphertext Letter: **E**

Character 6: 'T'

Plaintext Value ()P19

Calculation: $E(19)=(19+4) \bmod 26 = 23 \bmod 26$

Ciphertext Letter: **X**

Character 7: 'N'

Plaintext Value ()P13

Calculation: $E(13)=(13+4) \bmod 26 = 17 \bmod 26$

Ciphertext Letter: **R**

Character 8: 'O'

Plaintext Value ()P14

Calculation: $E(14)=(14+4) \bmod 26 = 18 \bmod 26$

Ciphertext Letter: **S**

Character 9: 'O'

Plaintext Value ()P14

Calculation: $E(14)=(14+4) \bmod 26 = 18 \bmod 26$

Ciphertext Letter: **S**

Character 10: 'N'

Plaintext Value ()P13

Calculation: $E(13)=(13+4) \bmod 26 = 17 \bmod 26$

Ciphertext Letter: **R**

Summary Table of Encryption Mapping

Original Plaintext	Numerical Map (P)	Shift Equation (P + 4)	Modulo 26 Result	Resulting Ciphertext
t	12	16	16	Q
M	4	8	8	I
E	4	8	8	I
E	19	23	23	X
T	-	-	-	Space
Space	0	4	4	E
A	19	23	23	X
T	-	-	-	Space
Space	13	17	17	R
N	14	18	18	S
O	14	18	18	S
O	13	17	17	R
N				

Following the deterministic execution of the shift protocol, the final encrypted output string that transmits across the untrusted communication vector is: "QIIX EX RSSR".

4. Authorized Retrieval and Decryption Mechanics

For an authorized member of the organization to access the original operational plan, they must implement a symmetric decryption protocol. Because the algorithm relies on a shared symmetric secret key, the receiver must possess pre-shared knowledge of both the transformation rule (Shift Cipher mechanism) and the operational key parameter ($K = 4$).

To reverse a shift of +4, the mathematical system dictates an additive inverse transformation, effectively shifting backward by 4 positions. Algebraically, this is modeled as:

$$D(C) = (C - 4) \bmod 26$$

If the subtraction produces a negative integer during algorithmic parsing, standard modular arithmetic rules dictate that the modulus (26) must be added repeatedly to the negative index until the value falls within the legitimate range [0, 25]. For example, if $C - 4 = -2$, the computation resolves as $-2 + 26 = 24$, mapping directly to 'Y'.

Granular Decryption Sequence:

- Decryption of 'Q' ($C = 16$): $D(16) = (16 - 4) \bmod 26 = 12 \rightarrow \mathbf{M}$
- Decryption of 'I' ($C = 8$): $D(8) = (8 - 4) \bmod 26 = 4 \rightarrow \mathbf{E}$
- Decryption of 'I' ($C = 8$): $D(8) = (8 - 4) \bmod 26 = 4 \rightarrow \mathbf{E}$
- Decryption of 'X' ($C = 23$): $D(23) = (23 - 4) \bmod 26 = 19 \rightarrow \mathbf{T}$
- Decryption of 'E' ($C = 4$): $D(4) = (4 - 4) \bmod 26 = 0 \rightarrow \mathbf{A}$
- Decryption of 'X' ($C = 23$): $D(23) = (23 - 4) \bmod 26 = 19 \rightarrow \mathbf{T}$
- Decryption of 'R' ($C = 17$): $D(17) = (17 - 4) \bmod 26 = 13 \rightarrow \mathbf{N}$
- Decryption of 'S' ($C = 18$): $D(18) = (18 - 4) \bmod 26 = 14 \rightarrow \mathbf{O}$
- Decryption of 'S' ($C = 18$): $D(18) = (18 - 4) \bmod 26 = 14 \rightarrow \mathbf{O}$
- Decryption of 'R' ($C = 17$): $D(17) = (17 - 4) \bmod 26 = 13 \rightarrow \mathbf{N}$

Through this automated mathematical translation, the authorized recipient successfully retrieves the underlying plaintext: "MEET AT NOON".

5. Cryptanalysis and Security Vulnerability Assessment

While simple to implement, the student organization's reliance on a fixed shift cipher represents a critical vulnerability in network security architecture. This cipher is acutely susceptible to two primary forms of cryptanalysis:

Brute Force (Exhaustive Key Search): The key space size of a standard shift cipher is exceptionally small. Because the English alphabet contains only 26 characters, there are exactly $26 - 1 = 25$ possible unique keys (a shift of 0 or 26 returns the original text completely unmodified). An adversary capturing "QIIX EX RSSR" can execute a complete exhaustive search in milliseconds manually, or microseconds programmatically, by iterating through keys 1 to 25 until legible text appears.

Frequency Analysis: In longer transmissions, monoalphabetic substitution fails entirely because it preserves the statistical properties of the natural language. In the English language, certain letters appear with highly predictable frequencies (e.g., 'E' is the most frequent at ~12.7%, followed by 'T', 'A', and 'O'). By counting the occurrences of characters within an extended ciphertext, an attacker can easily correlate the most frequent ciphertext characters with the known frequency profile of plaintext English. In this brief sample text, the double letters 'II' and 'SS' directly mirror the common linguistic frequency of double vowels and double consonants (like 'EE' and 'OO'). Therefore, this cipher fails to provide confidentiality against any semi-skilled adversary.

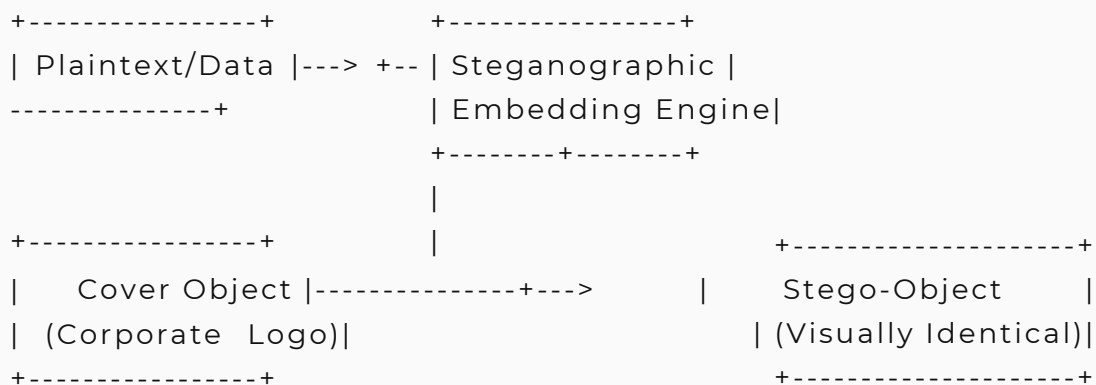
PART II: SUSPICIOUS DIGITAL ASSET INVESTIGATION (QUESTION 4)

1. Comprehensive Overview of Steganography

Steganography represents a highly specialized domain of network security and digital forensics focused on data hiding. Unlike traditional cryptography—which transforms a message into an unreadable string to conceal its *meaning*—steganography hides the very *existence* of the communication channel itself. While an encrypted file immediately signals to an interceptor that valuable data is being protected, a steganographically altered file disguises itself as an everyday digital artifact, escaping suspicion entirely.

In a digital forensic context, an asset like a corporate logo file (typically encoded as a .png, .bmp, or .jpeg) functions as the **cover object**. The hidden payload containing confidential information is embedded using specialized software tools, generating a structurally altered output known as the **stego-object**. Because a casual inspection reveals nothing out of the ordinary, investigators must utilize advanced forensic methodologies to uncover and extract the payload.

Data Embedding Protocol



2. Forensic Detection Methodologies (Steganalysis)

When digital investigators suspect that a corporate logo or media asset contains a hidden data cache, they apply the science of **steganalysis**. Because modern data hiding algorithms minimize visual degradation, detection relies on metadata auditing, spatial domain manipulation, and statistical anomalies.

Methodology A: Statistical Cryptanalysis and Color Histogram Deviation Analysis: Digital images store color values within complex pixel matrices. For example, in a standard 24-bit TrueColor image, each individual pixel contains three bytes representing the intensity channels for Red, Green, and Blue (RGB), yielding values between 0 and 255 per channel. When data hiding occurs, the specific values of these channels are altered systematically.

Investigators use automated toolsets to parse the image and generate precise color histograms mapping the distribution frequency of specific color values. A clean, unaltered corporate image possesses smooth, continuous transitions between colors, resulting in highly unified, natural curves on a frequency histogram. When an algorithm forcibly embeds binary fragments, it introduces unexpected, unnatural variations. The histogram displays jagged spikes and flat valleys—known as the "pairs of values" effect—where adjacent color parameters balance out artificially.

Forensic systems apply the Chi-Square (χ^2) statistical test to calculate whether the statistical distribution of the bits within the image deviates from the expected random distribution. A high probability match indicates that the pixel color values have been modified to match the uniform randomness typical of encrypted payloads.

Methodology B: Spatial Consistency Auditing and Least Significant Bit (LSB) Visual Filtering: The most ubiquitous approach to hiding data in images is LSB modification. This technique strips the lowest-order bit of a pixel's color byte and replaces it with a bit from the hidden payload. Because changing the last bit changes the color value by only 1 unit (e.g., from an RGB value of 140 to 141), the visual difference is completely imperceptible to the human eye.

Standard Pixel Byte (Value 140):	1 0 0 0 1 1 0
Modified Pixel Byte (Value 141):	0
	1 0 0 0 1 1 0 ^ [LSB replaced with Payload
Bit]	1

To detect this, forensic investigators apply LSB visual filtration. Tools like StegExpose or custom Python scripts strip away the most significant bits (which carry the core visual structure of the logo) and isolate only the least significant bits. If the image is clean, the isolated LSB layer looks like natural, unstructured random noise. If the logo contains hidden data, the LSB filtration reveals distinctive blocks or high-contrast structural patches where data was sequentially injected, exposing a non-random distribution of structural binary data.

Methodology C: File Structure and Metadata Integrity Inspection: Many rudimentary data-hiding applications do not inject data directly into the pixel configurations; instead, they append payloads to the file's structural boundaries. Every file format has designated hex markers establishing its boundary rules. A standard PNG file must initiate with the hex header sequence 89 50 4E 47 0D 0A 1A 0A and conclude with the End-of-Image trailer marker 49 45 4E 44 AE 42 60 82. A standard JPEG starts with FF D8 and terminates with FF D9.

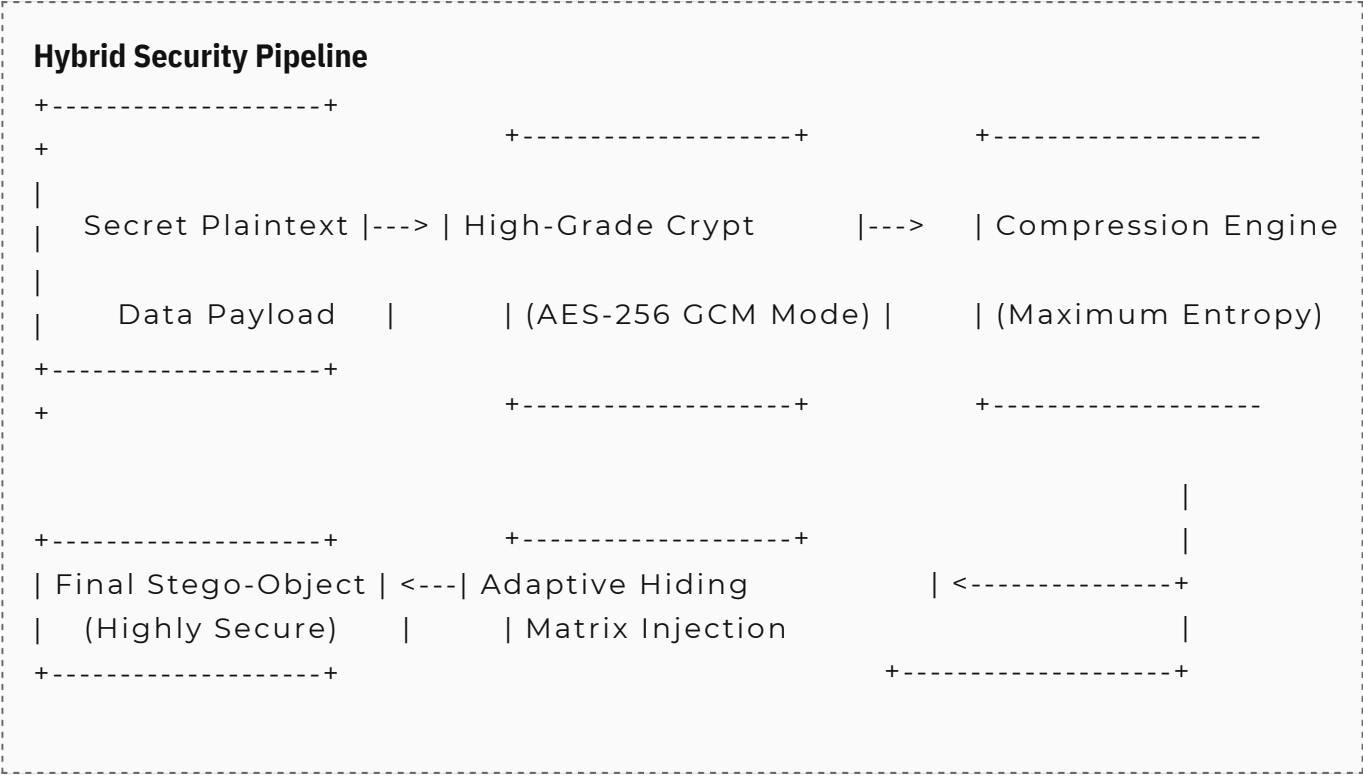
Forensic software parses the target logo file using hexadecimal analysis. If data has been appended past the structural end marker, the file size will be significantly larger than the dimension specifications

dictate, while the image displays normally. Investigators use automated tools to scrape any data located past the official end-of-file trailer, exposing hidden archives (like .zip or .txt objects) attached to the logo.

3. Countermeasures: Advanced Security Techniques to Strengthen Hidden Communications

If the objective shifts from detecting to protecting hidden operational logs, basic LSB replacement is insufficient. To shield sensitive transmissions from detection by forensic analysts, practitioners utilize advanced architectural frameworks that harden the stego-object against extraction and statistical analysis.

Technique 1: Implementation of a Double-Layer Hybrid Security Framework (Pre-Encryption and Compression): An unencrypted payload injected into an asset leaves a clear forensic trail. If an analyst extracts the payload, the underlying strategy is completely compromised. To prevent this single point of failure, a hybrid mechanism combining cryptography and steganography must be implemented.



Before embedding, the data payload is encrypted using a modern symmetric standard, such as Advanced Encryption Standard (AES) with a 256-bit key operating in Galois/Counter Mode (GCM). This ensures that even if a forensic investigator discovers and extracts the data layer, it appears as absolute cryptographic noise. Without the private AES key, it cannot be decrypted. The encrypted text is then passed through a compression algorithm (like DEFLATE). Compression strips away structural repetitions and patterns, maximizing the data's entropy. This makes it much harder for statistical steganalysis tools (like the Chi-Square test) to differentiate between the hidden data and normal pixel noise.

Technique 2: Adaptive Edge-Based and Pseudo-Random Matrix Hiding Mechanisms: Standard steganographic software embeds data sequentially, beginning at pixel (0,0) and proceeding systematically across the image grid. This creates an easily detectable boundary line that forensic scans instantly flag. To prevent this, data injection must adapt to the image's layout.

Instead of modifying pixels in order, the system uses a Pseudo-Random Number Generator (PRNG) driven by a shared secret seed key. This seed distributes the payload bits across non-consecutive pixels throughout the entire image. An analyst scanning the data sequentially will find nothing but disjointed fragments. Furthermore, the framework applies Adaptive Edge-Based Hiding. Statistical analysis tools detect changes much faster in smooth, flat areas of an image (like a clear blue sky). Edge-based algorithms scan the logo for high-contrast boundaries and complex textures—such as text edges, sharp geometric lines, or intricate patterns. Because these areas naturally contain highly volatile color transitions, changes to their pixel values blend in seamlessly, keeping the image's overall color distribution within normal thresholds.

4. Comparative Structural Analysis: Cryptography vs. Steganography

To conclude this forensic investigation framework, it is crucial to analyze the architectural differences between these two complementary pillars of information security.

Security Characteristic	Cryptography Domain	Steganography Domain
Primary Security Objective	To conceal the meaning and contents of the message being sent.	To conceal the very existence of the communication channel.
Visibility Classification	High Visibility. The output is obviously scrambled and flags attention.	Hidden Visibility. The output appears as an innocuous everyday object.
Mathematical Structure	Relies on modular arithmetic, prime factorization, and substitution blocks.	Relies on spatial domain modification, frequency transforms, and noise substitution.
Detection Status	Adversaries can easily identify that encryption is being used, even if they can't read it.	Adversaries remain completely unaware that communication is occurring unless they use deep forensics.
Failure Vulnerability	Susceptible to brute-force attacks, key leaks, and quantum processing acceleration.	Susceptible to file modification, conversion (e.g., JPEG to PNG), and statistical modeling.
Combined Integration	Acts as the foundational data preparation layer before hiding occurs.	Acts as the ultimate containment envelope to carry the encrypted data safely.

Through this comprehensive evaluation, it is clear that while cryptography protects the text from being read, steganography protects the sender and receiver from being targeted in the first place. When used together, they create a highly resilient, multi-layered security framework that can withstand modern digital forensic scrutiny.